HILBERT–SAMUEL MULTIPLICITY EQUALS THE KOSZUL EULER CHARACTERISTIC

THE STACKS PROJECT AUTHORS

Editorial note. The historical review incorrectly paraphrases equality of Euler sums as E_infinity=E_1. The source 1064–1069 only equates alternating length sums; no degeneration at E_1 is assumed.

Editorial note. The finite-difference identity is ordinary polynomial algebra and is an allowed prerequisite; it is not another counted problem.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian local algebra, Hilbert–Samuel polynomial, Artin–Rees, filtered-complex spectral sequences, length additivity, and the elementary finite-difference identity. None states the Koszul multiplicity equality.

Source reference. Original lemma 0663: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7922–7922. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 012K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1402–1402. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 012V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1413–1413. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00IO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 686–686. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 00K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 738–738. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AZY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13841–13841. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 07DU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9424–9424. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 00K8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 742–742. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00L8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 778–778. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1. [1, Theorem 1 in part B of Chapter IV] *Let A be a Noetherian local ring. Let $I = (f_1, \ldots, f_r) \subset A$ be an ideal of definition. Let M be a finite A-module. Then*

$$e_I(M, r) = \sum (-1)^i length_A H_i(K_\bullet(f_1, \ldots, f_r) \otimes_A M)$$

1. Algebraic multiplicities

Let $(A, \mathfrak{m}, \kappa)$ be a Noetherian local ring. Let M be a finite A-module and let $I \subset A$ be an ideal of definition (Algebra, Definition 07DU). Recall that the function

$$\chi_{I,M}(n) = \text{length}_A(M/I^n M) = \sum\nolimits_{p=0,\ldots,n-1} \text{length}_A(I^p M/I^{p+1} M)$$

is a numerical polynomial (Algebra, Proposition 00K8). The degree of this polynomial is equal to $\dim(\text{Supp}(M))$ by Algebra, Lemma 00L8.

Definition 1.1. In the situation above, assume $d \geq \dim(\text{Supp}(M))$. In this case, if $d > \dim(\text{Supp}(M))$, then we set $e_I(M, d) = 0$ and if $d = \dim(\text{Supp}(M))$, then we set $e_I(M, d)$ equal to $d!$ times the leading coefficient of the numerical polynomial $\chi_{I,M}$. Thus in both cases we have

$$\chi_{I,M}(n) \sim e_I(M, d)\frac{n^d}{d!} + \text{lower order terms}$$

The *multiplicity of M for the ideal of definition I* is $e_I(M) = e_I(M, \dim(\text{Supp}(M)))$.

Proof. Let us change the Koszul complex $K_\bullet(f_1, \ldots, f_r)$ into a cochain complex $K^\bullet$ by setting $K^n = K_{-n}(f_1, \ldots, f_r)$. Then $K^\bullet$ is sitting in degrees $-r, \ldots, 0$ and $H^i(K^\bullet \otimes_A M) = H_{-i}(K_\bullet(f_1, \ldots, f_r) \otimes_A M)$. The statement of the theorem makes sense as the modules $H^i(K^\bullet \otimes M)$ are annihilated by $f_1, \ldots, f_r$ (More on Algebra, Lemma 0663) hence have finite length. Define a filtration on the complex $K^\bullet$ by setting

$$F^p(K^n \otimes_A M) = I^{\max(0, p+n)}(K^n \otimes_A M), \quad p \in \mathbf{Z}$$

Since $f_i I^p \subset I^{p+1}$ this is a filtration by subcomplexes. Thus we have a filtered complex and we obtain a spectral sequence, see Homology, Section 012K. We have

$$E_0 = \bigoplus\nolimits_{p,q} E_0^{p,q} = \bigoplus\nolimits_{p,q} \text{gr}^p(K^{p+q} \otimes_A M) = \text{Gr}_I(K^\bullet \otimes_A M)$$

Since K^n is finite free we have

$$\mathrm{Gr}_I(K^\bullet \otimes_A M) = \mathrm{Gr}_I(K^\bullet) \otimes_{\mathrm{Gr}_I(A)} \mathrm{Gr}_I(M)$$

Note that $\mathrm{Gr}_I(K^\bullet)$ is the Koszul complex over $\mathrm{Gr}_I(A)$ on the elements $\overline{f}_1, \ldots, \overline{f}_r \in I/I^2$. A simple calculation (omitted) shows that the differential d_0 on E_0 agrees with the differential coming from the Koszul complex. Since $\mathrm{Gr}_I(M)$ is a finite $\mathrm{Gr}_I(A)$-module and since $\mathrm{Gr}_I(A)$ is Noetherian (as a quotient of $A/I[x_1, \ldots, x_r]$ with $x_i \mapsto \overline{f}_i$), the cohomology module $E_1 = \bigoplus E_1^{p,q}$ is a finite $\mathrm{Gr}_I(A)$-module. However, as above E_1 is annihilated by $\overline{f}_1, \ldots, \overline{f}_r$. We conclude E_1 has finite length. In particular we find that $\mathrm{Gr}_F^p(K^\bullet \otimes M)$ is acyclic for $p \gg 0$.

Next, we check that the spectral sequence above converges using Homology, Lemma 012V. The required equalities follow easily from the Artin-Rees lemma in the form stated in Algebra, Lemma 00IO. Thus we see that

$$\begin{aligned}\sum (-1)^i \mathrm{length}_A(H^i(K^\bullet \otimes_A M)) &= \sum (-1)^{p+q} \mathrm{length}_A(E_\infty^{p,q}) \\ &= \sum (-1)^{p+q} \mathrm{length}_A(E_1^{p,q})\end{aligned}$$

because as we've seen above the length of E_1 is finite (of course this uses additivity of lengths). Pick t so large that $\mathrm{Gr}_F^p(K^\bullet \otimes M)$ is acyclic for $p \geq t$ (see above). Using additivity again we see that

$$\sum (-1)^{p+q} \mathrm{length}_A(E_1^{p,q}) = \sum\nolimits_n \sum\nolimits_{p \leq t} (-1)^n \mathrm{length}_A(\mathrm{gr}^p(K^n \otimes_A M))$$

This is equal to

$$\sum\nolimits_{n=-r,\ldots,0} (-1)^n \binom{r}{|n|} \chi_{I,M}(t+n)$$

by our choice of filtration above and the definition of $\chi_{I,M}$ in Algebra, Section 00K4. The lemma follows from Lemma 0AZY and the definition of $e_I(M, r)$. □

References

[1] Jean-Pierre Serre. *Algèbre locale. Multiplicités*, volume 11 of *Cours au Collège de France, 1957–1958, rédigé par Pierre Gabriel. Seconde édition, 1965. Lecture Notes in Mathematics.* Springer-Verlag, Berlin, 1965.